

MODULE *AIExperimentationPlatform*

Three loop *AI* experimentation platform.
L1 experimentation propose, run, evaluate, commit on strict improve.
L2 durable execution append only *log*, workers crash and are reassigned.
L3 aiops budget burns on crash, stall burns on revert,
 breach triggers remediation which restores or
 halts.

EXTENDS *Naturals*, *Sequences*, *FiniteSets*

CONSTANTS *Workers*, *MaxExperiments*, *InitialBudget*, *StallCap*, *MaxMetric*

ASSUME

$\wedge \text{IsFiniteSet}(\text{Workers}) \wedge \text{Workers} \neq \{\}$
 $\wedge \{\text{MaxExperiments}, \text{InitialBudget}, \text{StallCap}, \text{MaxMetric}\} \subseteq \text{Nat}$
 $\wedge \text{MaxExperiments} > 0$

$\text{NoExp} \triangleq 0$
 $\text{ExpIds} \triangleq 1 \dots \text{MaxExperiments}$
 $\text{ExpS} \triangleq \{\text{"none"}, \text{"proposed"}, \text{"running"}, \text{"evaluated"}, \text{"committed"}, \text{"reverted"}\}$
 $\text{WS} \triangleq \{\text{"idle"}, \text{"busy"}, \text{"crashed"}\}$
 $\text{Auto} \triangleq \{\text{"ok"}, \text{"remediating"}, \text{"halted"}\}$
 $\text{EvK} \triangleq \{\text{"propose"}, \text{"start"}, \text{"finish"}, \text{"commit"}, \text{"revert"},$
 $\quad \text{"crash"}, \text{"reassign"}, \text{"downgrade"}, \text{"restore"}, \text{"halt"}\}$
 $\text{Event} \triangleq [\text{kind} : \text{EvK}, \text{exp} : \text{ExpIds} \cup \{\text{NoExp}\}]$

VARIABLES *exps*, *nextId*, *best*, *log*, *assignment*, *worker*, *budget*, *stall*, *autonomy*

vars $\triangleq \langle \text{exps}, \text{nextId}, \text{best}, \text{log}, \text{assignment}, \text{worker}, \text{budget}, \text{stall}, \text{autonomy} \rangle$

$\text{TypeOK} \triangleq$
 $\wedge \text{exps} \in [\text{ExpIds} \rightarrow [\text{state} : \text{ExpS}, \text{metric} : 0 \dots \text{MaxMetric}]]$
 $\wedge \text{nextId} \in 1 \dots \text{MaxExperiments} + 1$
 $\wedge \text{best} \in 0 \dots \text{MaxMetric}$
 $\wedge \text{log} \in \text{Seq}(\text{Event})$
 $\wedge \text{assignment} \in [\text{Workers} \rightarrow \text{ExpIds} \cup \{\text{NoExp}\}]$
 $\wedge \text{worker} \in [\text{Workers} \rightarrow \text{WS}]$
 $\wedge \text{budget} \in 0 \dots \text{InitialBudget}$
 $\wedge \text{stall} \in \text{Nat}$
 $\wedge \text{autonomy} \in \text{Auto}$

$\text{Init} \triangleq$
 $\wedge \text{exps} = [e \in \text{ExpIds} \mapsto [\text{state} \mapsto \text{"none"}, \text{metric} \mapsto 0]]$
 $\wedge \text{nextId} = 1$
 $\wedge \text{best} = 0$
 $\wedge \text{log} = \langle \rangle$
 $\wedge \text{assignment} = [w \in \text{Workers} \mapsto \text{NoExp}]$

$\wedge \text{worker} = [w \in \text{Workers} \mapsto \text{"idle"}]$
 $\wedge \text{budget} = \text{InitialBudget}$
 $\wedge \text{stall} = 0$
 $\wedge \text{autonomy} = \text{"ok"}$

$\text{Emit}(k, e) \triangleq \log' = \log \circ \langle [kind \mapsto k, \text{exp} \mapsto e] \rangle$

$\mathbb{L}1$ experimentation

$\text{Propose} \triangleq$

$\wedge \text{autonomy} = \text{"ok"}$
 $\wedge \text{nextId} \in \text{ExpIds}$
 $\wedge \text{exps}' = [\text{exps} \text{ EXCEPT } ![nextId] = [state \mapsto \text{"proposed"}, metric \mapsto 0]]$
 $\wedge \text{nextId}' = \text{nextId} + 1$
 $\wedge \text{Emit}(\text{"propose"}, \text{nextId})$
 $\wedge \text{UNCHANGED } \langle \text{best}, \text{assignment}, \text{worker}, \text{budget}, \text{stall}, \text{autonomy} \rangle$

$\text{Start}(e, w) \triangleq$

$\wedge \text{autonomy} \neq \text{"halted"}$
 $\wedge \text{exps}[e].state = \text{"proposed"}$
 $\wedge \text{worker}[w] = \text{"idle"}$
 $\wedge \text{exps}' = [\text{exps} \text{ EXCEPT } ![e].state = \text{"running"}]$
 $\wedge \text{assignment}' = [\text{assignment} \text{ EXCEPT } ![w] = e]$
 $\wedge \text{worker}' = [\text{worker} \text{ EXCEPT } ![w] = \text{"busy"}]$
 $\wedge \text{Emit}(\text{"start"}, e)$
 $\wedge \text{UNCHANGED } \langle \text{nextId}, \text{best}, \text{budget}, \text{stall}, \text{autonomy} \rangle$

$\text{Finish}(e, w, m) \triangleq$

$\wedge \text{worker}[w] = \text{"busy"} \wedge \text{assignment}[w] = e$
 $\wedge \text{exps}[e].state = \text{"running"}$
 $\wedge \text{exps}' = [\text{exps} \text{ EXCEPT } ![e] = [state \mapsto \text{"evaluated"}, metric \mapsto m]]$
 $\wedge \text{assignment}' = [\text{assignment} \text{ EXCEPT } ![w] = \text{NoExp}]$
 $\wedge \text{worker}' = [\text{worker} \text{ EXCEPT } ![w] = \text{"idle"}]$
 $\wedge \text{Emit}(\text{"finish"}, e)$
 $\wedge \text{UNCHANGED } \langle \text{nextId}, \text{best}, \text{budget}, \text{stall}, \text{autonomy} \rangle$

$\text{Commit}(e) \triangleq$

$\wedge \text{exps}[e].state = \text{"evaluated"}$
 $\wedge \text{exps}[e].metric > \text{best}$
 $\wedge \text{exps}' = [\text{exps} \text{ EXCEPT } ![e].state = \text{"committed"}]$
 $\wedge \text{best}' = \text{exps}[e].metric$
 $\wedge \text{stall}' = 0$
 $\wedge \text{Emit}(\text{"commit"}, e)$
 $\wedge \text{UNCHANGED } \langle \text{nextId}, \text{assignment}, \text{worker}, \text{budget}, \text{autonomy} \rangle$

$\text{Revert}(e) \triangleq$

$\wedge \text{exps}[e].state = \text{"evaluated"}$

$\wedge \text{exps}[e].\text{metric} \leq \text{best}$
 $\wedge \text{exps}' = [\text{exps} \text{ EXCEPT } ![e].\text{state} = \text{"reverted"}]$
 $\wedge \text{stall}' = \text{stall} + 1$
 $\wedge \text{Emit}(\text{"revert"}, e)$
 $\wedge \text{UNCHANGED } \langle \text{nextId}, \text{best}, \text{assignment}, \text{worker}, \text{budget}, \text{autonomy} \rangle$

L2 durable execution

$\text{Crash}(w) \triangleq$
 $\wedge \text{worker}[w] = \text{"busy"}$
 $\wedge \text{worker}' = [\text{worker} \text{ EXCEPT } ![w] = \text{"crashed"}]$
 $\wedge \text{budget}' = \text{IF } \text{budget} > 0 \text{ THEN } \text{budget} - 1 \text{ ELSE } 0$
 $\wedge \text{Emit}(\text{"crash"}, \text{assignment}[w])$
 $\wedge \text{UNCHANGED } \langle \text{exps}, \text{nextId}, \text{best}, \text{assignment}, \text{stall}, \text{autonomy} \rangle$

$\text{Reassign}(w) \triangleq$
 $\wedge \text{worker}[w] = \text{"crashed"}$
 $\wedge \text{LET } e \triangleq \text{assignment}[w] \text{ IN}$
 $\quad \wedge \text{exps}' = [\text{exps} \text{ EXCEPT } ![e].\text{state} = \text{"proposed"}]$
 $\quad \wedge \text{assignment}' = [\text{assignment} \text{ EXCEPT } ![w] = \text{NoExp}]$
 $\quad \wedge \text{worker}' = [\text{worker} \text{ EXCEPT } ![w] = \text{"idle"}]$
 $\quad \wedge \text{Emit}(\text{"reassign"}, e)$
 $\wedge \text{UNCHANGED } \langle \text{nextId}, \text{best}, \text{budget}, \text{stall}, \text{autonomy} \rangle$

L3 aiops

$\text{Breach} \triangleq \text{budget} = 0 \vee \text{stall} \geq \text{StallCap}$

$\text{Downgrade} \triangleq$
 $\wedge \text{autonomy} = \text{"ok"}$
 $\wedge \text{Breach}$
 $\wedge \text{autonomy}' = \text{"remediating"}$
 $\wedge \text{Emit}(\text{"downgrade"}, \text{NoExp})$
 $\wedge \text{UNCHANGED } \langle \text{exps}, \text{nextId}, \text{best}, \text{assignment}, \text{worker}, \text{budget}, \text{stall} \rangle$

$\text{Restore} \triangleq$
 $\wedge \text{autonomy} = \text{"remediating"}$
 $\wedge \text{autonomy}' = \text{"ok"}$
 $\wedge \text{budget}' = \text{InitialBudget}$
 $\wedge \text{stall}' = 0$
 $\wedge \text{Emit}(\text{"restore"}, \text{NoExp})$
 $\wedge \text{UNCHANGED } \langle \text{exps}, \text{nextId}, \text{best}, \text{assignment}, \text{worker} \rangle$

$\text{Halt} \triangleq$
 $\wedge \text{autonomy} = \text{"remediating"}$
 $\wedge \text{autonomy}' = \text{"halted"}$
 $\wedge \text{Emit}(\text{"halt"}, \text{NoExp})$

$$\wedge \text{UNCHANGED } \langle \text{exps}, \text{nextId}, \text{best}, \text{assignment}, \text{worker}, \text{budget}, \text{stall} \rangle$$

$$\text{Next} \triangleq$$

$$\begin{aligned} &\vee \text{Propose} \\ &\vee \exists e \in \text{ExpIds}, w \in \text{Workers} : \text{Start}(e, w) \\ &\vee \exists e \in \text{ExpIds}, w \in \text{Workers}, m \in 0 \dots \text{MaxMetric} : \text{Finish}(e, w, m) \\ &\vee \exists e \in \text{ExpIds} : \text{Commit}(e) \\ &\vee \exists e \in \text{ExpIds} : \text{Revert}(e) \\ &\vee \exists w \in \text{Workers} : \text{Crash}(w) \\ &\vee \exists w \in \text{Workers} : \text{Reassign}(w) \\ &\vee \text{Downgrade} \vee \text{Restore} \vee \text{Halt} \end{aligned}$$

$$\text{Fairness} \triangleq$$

$$\begin{aligned} &\wedge \text{SF}_{\text{vars}}(\exists e \in \text{ExpIds}, w \in \text{Workers} : \text{Start}(e, w)) \\ &\wedge \text{SF}_{\text{vars}}(\exists e \in \text{ExpIds}, w \in \text{Workers}, m \in 0 \dots \text{MaxMetric} : \text{Finish}(e, w, m)) \\ &\wedge \text{WF}_{\text{vars}}(\exists e \in \text{ExpIds} : \text{Commit}(e) \vee \text{Revert}(e)) \\ &\wedge \text{WF}_{\text{vars}}(\exists w \in \text{Workers} : \text{Reassign}(w)) \\ &\wedge \text{WF}_{\text{vars}}(\text{Downgrade}) \\ &\wedge \text{WF}_{\text{vars}}(\text{Restore} \vee \text{Halt}) \end{aligned}$$

$$\text{Spec} \triangleq \text{Init} \wedge \Box[\text{Next}]_{\text{vars}} \wedge \text{Fairness}$$

properties

$$\text{RatchetMonotone} \triangleq \Box[\text{best}' \geq \text{best}]_{\text{vars}}$$

$$\text{LogAppendOnly} \triangleq$$

$$\Box[\text{Len}(\log') \geq \text{Len}(\log) \wedge \text{SubSeq}(\log', 1, \text{Len}(\log)) = \log]_{\text{vars}}$$

$$\text{Terminal}(s) \triangleq s \in \{ \text{"committed"}, \text{"reverted"} \}$$

$$\text{ProposedTerminates} \triangleq \forall e \in \text{ExpIds} :$$

$$(\text{exps}[e].\text{state} = \text{"proposed"}) \rightsquigarrow \text{Terminal}(\text{exps}[e].\text{state})$$

$$\text{BreachHandled} \triangleq \text{Breach} \rightsquigarrow (\neg \text{Breach} \vee \text{autonomy} = \text{"halted"})$$